

Sampling

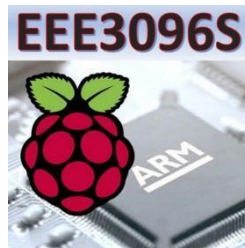
Sampling & Data Conversion · Module 4, Lecture 1 of 4

Travimadox Webb^{1,2}

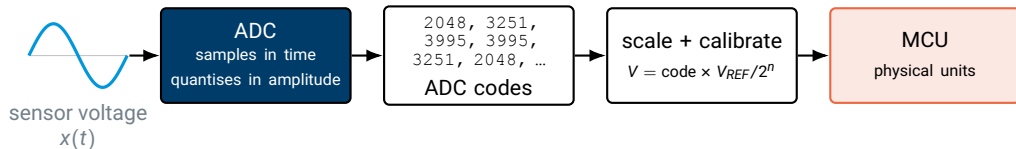
September 20, 2026

¹ Department of Electrical Engineering
University of Cape Town

² African Robotics Unit (ARU)



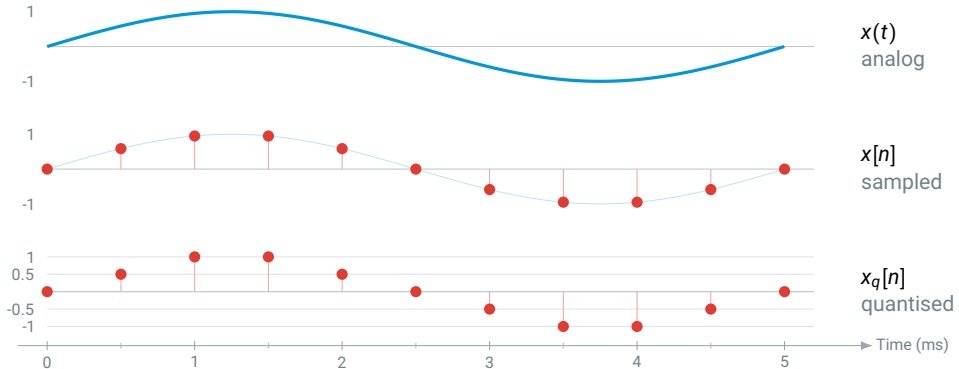
Why does a microcontroller need samples?



Communications moves numbers. Data conversion connects those numbers to physical signals.

An ADC does two separate things: it **samples** in time and **quantises** in amplitude. Today is about sampling.

Sampling selects times; quantisation selects values



Sampling changes time. Quantisation changes amplitude.

Illustrative 0.5 V grid, not a bit depth: $0.588 \rightarrow 0.5$, $0.951 \rightarrow 1.0$.
Stems mark sample instants and heights; they are not a voltage trajectory.



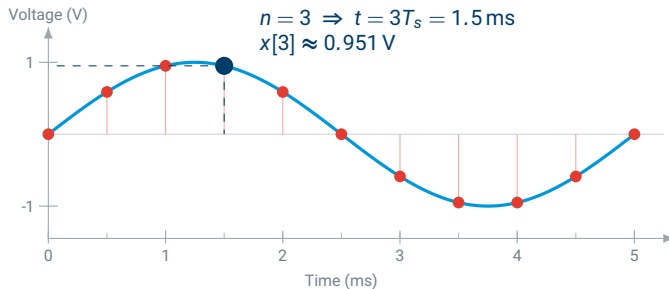
Each sample has a time and a value

$$x[n] = x(nT_s)$$

$$T_s = 1/f_s$$

n integer sample index
 T_s time between samples
 $x[n]$ value at $t = nT_s$

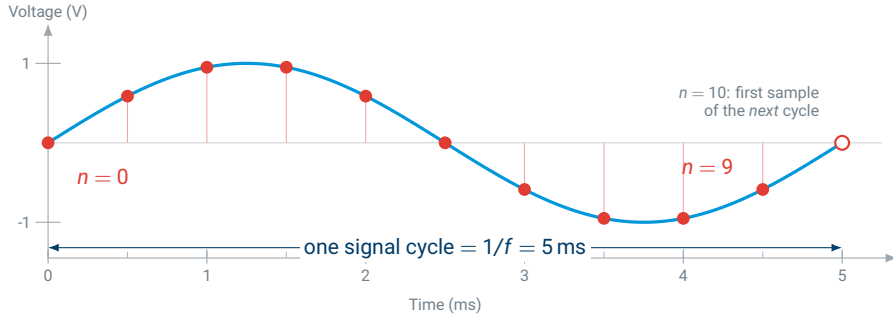
Here $f_s = 2000$ samples/s, so
 $T_s = 0.5$ ms.



The index identifies the sample; multiplying by T_s gives its time.



Signal frequency and sampling frequency are different

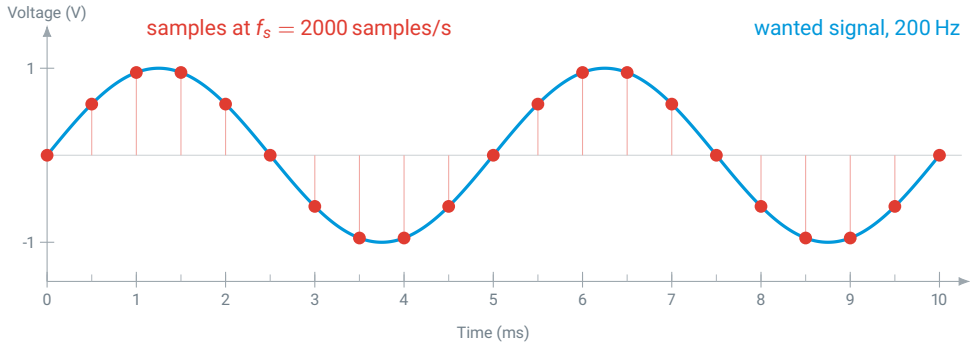


$$f = 200 \text{ cycles/s} \quad f_s = 2000 \text{ samples/s} \quad \frac{f_s}{f} = 10 \text{ samples per cycle}$$

Signal frequency describes the waveform. Sampling frequency describes the measurements.



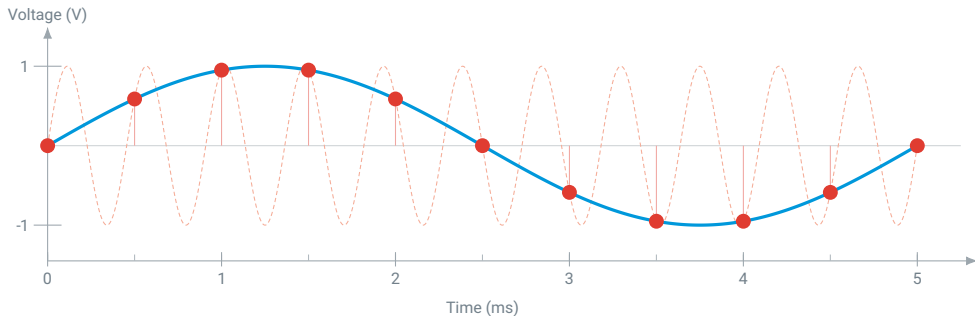
Ten samples per cycle capture the 200 Hz pattern



Could a different analog signal pass through these same points?

Different signals can produce identical samples

— wanted 200 Hz — unwanted 2200 Hz • shared samples



The samples alone cannot distinguish these two signals.

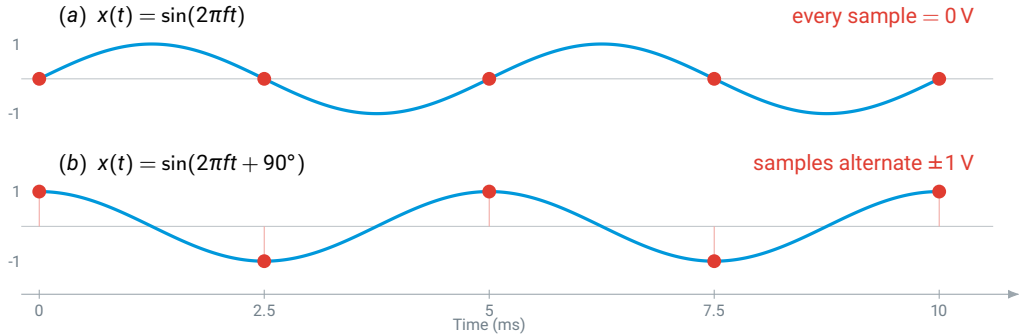
At $f_s = 2000$ samples/s, 2200 Hz aliases to 200 Hz.



In this example the apparent frequency is $2200 - 2000 = 200$ Hz; that subtraction is not a general formula.

Exactly two samples per cycle can miss the sine

FOR THIS SLIDE ONLY: $f_s = 400$ samples/s, $T_s = 2.5$ ms



At exactly two samples per cycle, the observed samples depend critically on phase.



The sampling theorem gives the reconstruction condition

$$f_s > 2f_{max}$$

A signal bandlimited to f_{max} can be reconstructed from ideal samples when $f_s > 2f_{max}$.

Bandlimited

No frequency components above f_{max} .

Two names that are easy to swap

Nyquist rate $2f_{max}$, a signal property

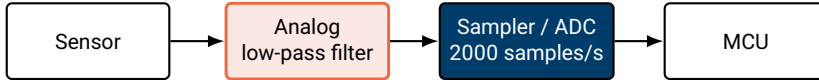
Nyquist frequency $f_s/2$, a sampler property

For the wanted 200 Hz signal: Nyquist rate = 400 samples/s. At $f_s = 2000$ samples/s the Nyquist frequency is 1000 Hz.

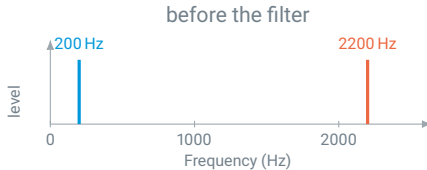


Filter unwanted content before sampling

wanted content: up to 200 Hz



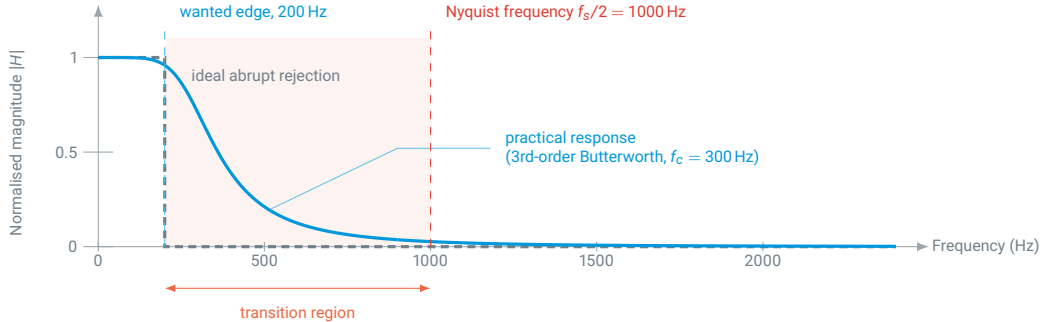
example unwanted component: 2200 Hz



Suppress the unwanted 2200 Hz component before it can appear as 200 Hz.



Real filters need room to roll off



A higher sampling rate leaves more room between the wanted band and the Nyquist frequency.



Choose the sample rate and filter together

A vibration signal carries wanted content up to 200 Hz. What sampling rate should we choose?

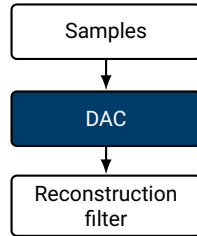
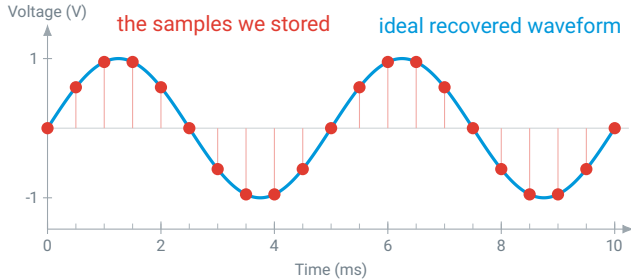
- 1 Wanted band: 0 to 200 Hz, so $f_{max} = 200$ Hz.
- 2 Ideal sampling requirement: $f_s > 2f_{max} = 400$ samples/s.
- 3 Candidate sampling rate: $f_s = 2000$ samples/s.
- 4 Verify passband accuracy to 200 Hz and rejection of unwanted content.

That candidate gives **10 samples per 200 Hz cycle** and a **Nyquist frequency of 1000 Hz**.

2000 samples/s is a candidate design choice; filter performance must still be checked.



What does reconstruction mean?



Reconstruction uses the samples and a bandwidth assumption to recover a waveform between sample times.

The smooth curve above is the known 200 Hz generating sine, drawn as the ideal result. Joining samples with straight lines is not exact reconstruction.

Check your understanding

- a. Does sampling round 0.588 V to 0.5 V?
- b. At 2000 samples/s, why can 2200 Hz look like 200 Hz?
- c. Can a digital low-pass filter separate a 2200 Hz component that has aliased to 200 Hz from a genuine 200 Hz signal?



Check your understanding

- a. Does sampling round 0.588 V to 0.5 V?

No. That is quantisation.

- b. At 2000 samples/s, why can 2200 Hz look like 200 Hz?

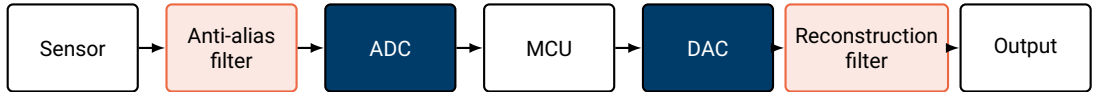
These signals can have identical values at the sampling instants.

- c. Can a digital low-pass filter separate a 2200 Hz component that has aliased to 200 Hz from a genuine 200 Hz signal?

Not from the sampled data alone. Suppress it before sampling.



Sampling determines which time variation we can preserve



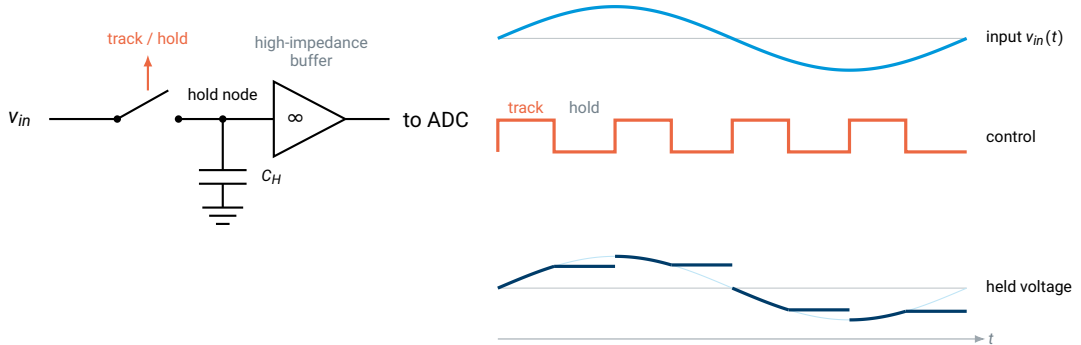
- Sampling selects measurement times.
- Suitable bandwidth control and sampling prevent unwanted aliasing.
- Quantisation sets the available amplitude levels.

A microcontroller needs samples because that is the only form in which it can hold a physical signal; the rate you choose decides which time variation survives.

Next: how ADCs and DACs represent amplitude, and why resolution and accuracy differ.



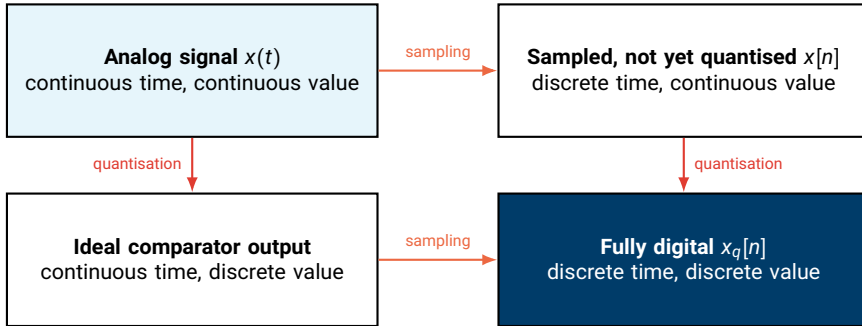
Backup: sample-and-hold, a conceptual function



Track: the capacitor follows the input. **Hold:** the stored voltage is retained for conversion.

The held voltage is a physical waveform defined at every instant between sampling times, so it is not itself a discrete-time sequence, and its staircase shape does not imply amplitude quantisation. In modern parts this function is usually integrated inside the sampling ADC rather than built as a separate circuit.

Backup: continuous and discrete are two independent axes



Sampling moves you across the *time* classification. Quantisation moves you across the *amplitude* classification. The two are independent.

Backup: the Shannon–Hartley channel-capacity bound

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

A different question

The sampling theorem bounds the *sampling rate* needed for a given signal bandwidth. Shannon–Hartley bounds the *information rate* C in bits/s that a channel of bandwidth B can carry at signal-to-noise ratio S/N . The two are often conflated because both involve a bandwidth.

A 3 kHz channel at 30 dB SNR ($S/N = 1000$) gives $C = 3000 \log_2(1001) \approx 29\,900$ bits/s. This is a bound on that idealised channel, not a description of any particular modem standard.

